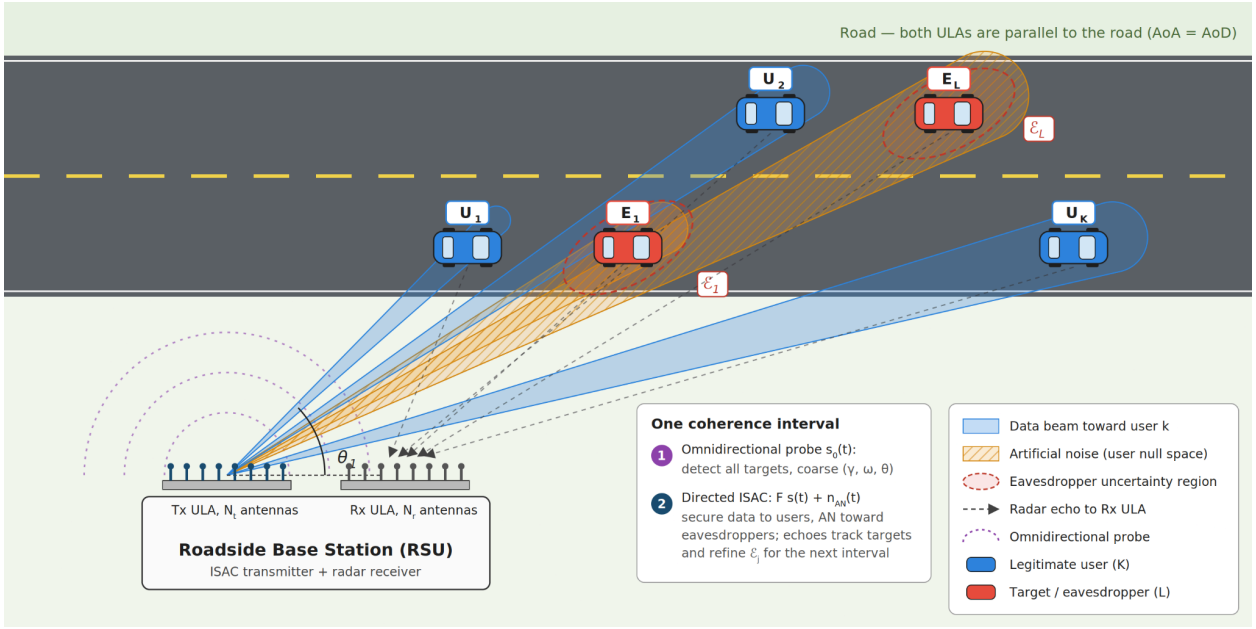
Sum-Secrecy Rate Maximization in OTFS-Assisted ISAC

I. System Model



We have a roadside Base Station (BS) serving K communication users belonging to the set and indexed as $K = \{k_1, k_2, k_3, \ldots, k_K\}$ and J radar targets belonging to the set and indexed as $J = \{j_1, j_2, \ldots, j_J\}$ that are treated as potential eavesdroppers.
The sets K and J are disjoint, and the K communication users known to the BS are to be sent confidential data to be kept secret from the J eavesdroppers, which are to be discovered and localised by the BS through sensing.
The BS is equipped with a transmit ULA with N_t transmit antennas and a receive ULA with N_r receive antennas which are separate and sufficiently spaced apart such that the self interference from the radar echo does not affect the transmit signal. Both arrays are parallel to the road, leading to the AoA coinciding with the AoD, indexed as θ_i for target i.

We operate the following steps in a coherence interval.

1. Isotropic sensing probe: Omnidirectional radar signal for detection and coarse delay-Doppler-angle estimation of all targets.

2. Directed ISAC transmission: The BS transmits confidential OTFS data to the K legitimate users via a directional precoding matrix, superimposed with artificial noise (AN) shaped to degrade any eavesdropper's effective channel. The reflected echoes from this same transmission are simultaneously used to track all targets and refine the eavesdropper location estimate for the next coherence interval.

II. OTFS Modulation

Sequence of bits mapped to N × M data symbols $x_i[l, k] \in \mathbb{C}$ in the 2D Delay-Doppler Domain, where
$0 \le l \le M - 1;\ 0 \le k \le N - 1$
N: No. of timeslots
M: No. of subcarriers
The DD-domain symbols are mapped to the time-frequency domain via the inverse symplectic finite Fourier transform (ISFFT)

$$X_k[n, m] = \frac{1}{\sqrt{MN}} \sum_{k'=0}^{N-1} \sum_{l=0}^{M-1} x_k[k', l]\, e^{j2\pi\left(\frac{nk'}{N} - \frac{ml}{M}\right)}$$

And the generated ISAC signal for user k via Heisenberg transform:

$$s_k(t) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} X_k[n, m]\, e^{j2\pi m\Delta f(t-nT)}\, g_{tx}(t - nT)$$

Where $g_{tx}(t)$ is the transmit filter, T is the symbol duration, Δf is the frequency spacing satisfying $T\Delta f = 1$

III. Radar Signal Model

Taking P = K ∪ J, such that we have p vehicles and targets,
The omnidirectional sensing signal $s_0(t)$ is sent with equal gain in all directions, no beamforming

We get the reflected signal from all i targets ∈ P at the receive array:

$$r^{(0)}(t) = \sum_{i=1}^{P} \beta_i \, b\left(\theta_i\right) s_0\left(t - \gamma_i\right) e^{j2\pi\omega_i t} + z(t)$$

$z(t) \in \mathbb{C}^{N_r \times 1}$ (AWGN)
And where β_i, γ_i, and ω_i are the reflection coefficient, round-trip delay, and round-trip Doppler shift of target i, respectively
$b(\theta_i) = [b_1(\theta_i), \ldots, b_{N_r}(\theta_i)]^T$ is the receive steering vector

We obtain an initial, coarse estimate of (γ_j, ω_j, θ_j) for every detected target, which we will refine into an uncertainty region

We then prepare a multi-beam ISAC signal with the confidential data of the K users:
$s(t) = [s_1(t), \ldots, s_K(t)]^T$
transmitted over the N_t antennas through a precoding matrix $F \in \mathbb{C}^{N \times K}$ whose k-th column steers $s_k(t)$ toward the predicted direction θ_k of user k

$$f_k = \sqrt{\frac{p_k}{N_t}}\, a\left(\theta^-_k\right)$$

And

$$a(\theta) = \left[1,\ e^{j\pi sin\theta},\ \ \ldots,\ e^{j\pi\left(N_t - 1\right)sin\theta}\right]^T$$

with p_k the power allocated to user k. To additionally degrade any eavesdropper's reception, the transmitted signal is superimposed with artificial noise $n(t) \sim CN(0, S)$, $S \succeq 0$,
giving the full transmitted signal

$$\tilde{s}(t) = F\, s(t) + n_{AN}(t)$$

From the BS's perspective, the radar sensing channel is both time and frequency selective

$$H(t,\tau) = \sum_{i=1}^{P} \beta_i \, b\left(\theta_i\right) a^H\left(\theta_i\right) \delta\left(\tau - \gamma_i\right) e^{j2\pi\omega_i t}$$

and the resulting N_r-dimensional received echo is

$$r(t) = \sum_{i=1}^{P} \beta_i \, b\left(\theta_i\right) a^H\left(\theta_i\right) \widetilde{s}\left(t - \gamma_i\right) e^{j2\pi\omega_i t} + z(t)$$

IV. Radar Measurement Model

We have asymptotically orthogonal receive steering vectors for different targets under favorable massive-MIMO condition

I.e. $b^H(\theta_i)\, b(\theta_{i'}) \to 0$ for i ≠ i′ as $N_r \to \infty$, so the BS can distinguish echoes reflected by different targets/vehicles

Matched filtering each separated echo against the known transmitted waveform $s_i(t)$ over DD search variables (γ, ω):

$$\tilde{r}_i(\gamma, \omega) = \int_0^{\Delta T} r_i(t)\, s_i^*(t - \gamma)\, e^{-j\omega t}\, dt$$

$$= \int_0^{\Delta T} \beta_i\, b(\theta_i)\, a^H(\theta_i)\, \tilde{s}(t - \gamma_i)\, s_i^*(t - \gamma)\, e^{-j(\omega - \omega_i)t}\, dt + z_i$$

With ΔT = NT the frame duration, yields a peak when (γ, ω) is tuned to target i's true (γ_i, ω_i), giving the delay and Doppler estimates $\hat{\gamma}_i$, $\hat{\omega}_i$ for every i ∈ P

Received signal for target i can be given by:

$$r_i = G_m\, \beta_i\, b(\theta_i)\, a^H(\theta_i)\, f_i + z_i$$

Where G_m is the SNR gain by matched filtering (identical to $s_i(t)$ energy)

And z_i ~ CN(0, σ^2I), i.i.d., is the filtered noise term

$$z_i = \int_0^{\Delta T} z(t)\, s_i^*(t - \gamma)\, e^{-j\omega t}\, dt$$

V. Communication Model (downlink)

a. For K users:

The downlink channel model is LoS dominated:

$$h_k^D(t) = \sqrt{\frac{c}{4\pi f_c d_k^2}}\, a^H\left(\theta_k\right) e^{j2\pi v_k t}$$

Where c is the speed of light, f_c is the carrier frequency, v_k the Doppler shift due to user k's motion, and d_k the distance between user k and the BS

The received signal at the i-th vehicle with the precoded data and AN would be:

$$y_k(t) = h_k e^{j2\pi v_k t}\, a^H\left(\theta_k\right)\left(f_k s_k(t) + \sum_{k' \neq k} f_{k'} s_{k'}(t) + n_{AN}(t)\right) + w(t)$$

Where w(t) is AWGN with PSD = N_0 and $h_k = \sqrt{\frac{c}{4\pi f_c d_k^2}}$

Post multi-carrier demodulation and receive filtering to obtain TF domain samples:

$$Y_i[n,m] = \int y_i(t)\, g_{rx}^*(t - n\Delta T)\, e^{-j2\pi m\Delta f(t-nT)}\, dt$$

Where g_{rx} is the receive filter
Finally, the TF domain samples are transformed to the DD domain via symplectic finite Fourier transform (SFFT):

$$y_k[u,l] = h_k\, a^H\left(\theta_k\right) f_k\, x_k\left[\left(u - u_k\right)_N, \left(l - l_k\right)_M\right] + w[u,l]$$

We consider sufficient doppler resolution such that $v_i = k_i/(NT)$ with $l_k = 0$

b. For J eavesdroppers:

An eavesdropper attempting to intercept the data stream intended for user k experiences a channel of the same structural form as the user, but with (θ_j, v_j, d_j) replacing (θ_k, v_k, d_k), and without knowledge of the precoder's exact alignment to it (the BS does not intentionally steer toward j).

The eavesdropper's received signal when attempting to decode user k's stream is

$$y_{j,k}[u,l] = h_j\, a^H(\theta_j)\, f_k\, x_k\left[\left(u - u_j\right)_N,\ \left(l - l_j\right)_M\right] + w_j[u,l]$$

where h_j, θ_j, v_j take their true (unknown) values, while the BS's design of F and S can only be based on the estimated region containing them.

VI. Sensing parameter estimation and Prediction

a. For K users:

Based on the received echoes (r_i), the BS is capable of inferring the angular parameters and the reflected coefficients. For the reflected coefficient β_i, it is calculated as

$$\beta_i = \frac{\xi}{2d_i}$$

where ξ represents the radar cross section (RCS)

Considering that the delay of echo γ_i can be expressed as the round-trip range distance dividing by the signal propagation speed, i.e., $\gamma_i = 2d_i/c$, and using the estimated delay $\hat{\gamma}_i$ obtained at the BS, the estimate of β_i can be easily obtained as $\hat{\beta}_i = \xi/(c\hat{\gamma}_i)$. Substituting $\hat{\beta}_i$ into the equation for r_i and after straightforward manipulations, we obtain:

$$r_i = \hat{\beta}_i G_m \sqrt{p_i} \left[1 \; e^{j\pi sin\theta_i} \; \vdots \; e^{j\pi\left(N_r - 1\right)sin\theta_i}\right] \sum_{i=1}^{N_t} e^{j\pi(i-1)\left(sin\theta^{-}_i - sin\theta_i\right)} + z_i$$

From the perspective of the optimal ML estimation, we aim at maximizing the likelihood function $p(r_i \mid \theta_i)$. Given the number of the transmit antennas N_t, we can represent the possible values of θ_i by a discrete set of N_t angles, denoted by $\Theta = \{\pi/N_t, 2\pi/N_t, \ldots, \pi\}$. Therefore, the maximization of the likelihood function is given by:

$$\hat{\theta}_i = arg \max_{\theta_i \in \Theta} \; p\left(r_i | \theta_i\right)$$

Instead of an exhaustive search in the set Θ is required to find the optimal estimate, to reduce the estimation complexity, we can choose only the values in the set Θ that are close to the predicted angle $\bar{\theta}_i$ to find the estimated angle $\hat{\theta}_i$ that maximizes the likelihood function $p(r_i \mid \theta_i)$

Let us denote the Cartesian coordinates of the BS and the i-th vehicle by $q_{BS} = [q_{x,BS}, q_{y,BS}]^T$ and $q_i = [q_{x,i}, q_{y,i}]^T$, where the subscripts x and y denote the coordinates on x-axis and y-axis, respectively. Given the estimates of the delay $\hat{\gamma}_i$ and the angle $\hat{\theta}_i$, we can estimate the location of vehicle i at the current instant

$$\hat{q}_{x,i} = q_{x,BS} + \frac{c\hat{\gamma}_i sin\hat{\theta}_i}{2}$$

$$\hat{q}_{y,i} = q_{y,BS} + \frac{c\hat{\gamma}_i cos\hat{\theta}_i}{2}$$

Furthermore, the speed for vehicle i can be estimated by using its relationship with the Doppler estimate, i.e

$$\hat{s}_i = \frac{c\,\hat{\omega}_i}{f_c cos\hat{\theta}_i}$$

For the OTFS frame duration of ΔT and the predicted location for the i-th vehicle in the following time instant can be expressed as:

$$\bar{q}_i = \hat{q}_i + \Delta T \cdot \hat{s}_i \cdot [1, 0]^T$$

Assuming negligible speed change over one coherence interval, the predicted angle for the next time instant is:

$$\theta_k^{new} = arcsin \frac{\bar{q}_{x,k} - q_{x,BS}}{\|\bar{q}_k - q_{BS}\|}$$

This predicted angle is used to form f_k in the ISAC transmission for the next frame

b. For eavesdroppers:

Eavesdroppers' location is bounded by an uncertainty region derived from estimation theory.

Let $\eta_j = [\gamma_j, \omega_j, \theta_j]^T$ denote the unknown delay-Doppler-angle parameters of eavesdropper j, estimated from the pilot omnidirectional echo.

The Fisher Information Matrix for η_j yields the Cramer-Rao Bound:

$$Cov\left(\hat{\eta}_j - \eta_j\right) \succcurlyeq J^{-1}\left(\eta_j\right) \triangleq C_j$$

which defines an ellipsoidal uncertainty region around the point estimate $\hat{\eta_j}$

$$E_j = \left\{\eta: \left(\eta - \hat{\eta}_j\right)^T C_j^{-1} \left(\eta - \hat{\eta}_j\right) \leq \epsilon\right\}$$

for a confidence parameter ϵ. As more echoes accumulate for the same detected target across frames, C_j can be refined (shrinking E_j), tightening the secrecy guarantee over time.

VII. Problem Formulation

a. Refining Artificial Noise

Nulling an entire uncertainty region costs more spatial degrees of freedom than nulling a point, and may be infeasible with a limited array. We therefore construct AN explicitly in the null space of every legitimate user's channel, so it degrades potential eavesdroppers while not affecting any user in K.

Artificial noise matrix formulation

b. Achievable Rates

SINR:

Thus the achievable rate is given by:

$$R_k = log_2\left(1 + SINR_k\right)$$

For an eavesdropper,

$$R_{j,k}^{Eve} = \max_{\eta_j \in E_j} log_2\left(1 + \frac{N_{j,k}}{D_{j,k}\left(\theta_j\right)}\right)$$

c. Sum Secrecy Rate Maximization

$$
\begin{aligned}
\max_{\{\mathbf{W}_k\},\mathbf{Q}} \quad & \sum_{k\in\mathcal{K}} \left[R_k - \max_{j\in\mathcal{J}} R_{j,k}^{\mathrm{Eve}} \right]^+ \\
\text{s.t.} \quad & \mathrm{tr}(\mathbf{R}_x) \le P_{\max} \\
& \mathrm{SINR}_k \ge 2^{R_{\min}} - 1, \ \forall k \in \mathcal{K} \\
& \mathbf{a}^H(\hat{\theta}_k)\mathbf{R}_x\mathbf{a}(\hat{\theta}_k) \ge \Gamma_{\mathrm{trk}}, \ \forall k \in \mathcal{K} \\
& \mathrm{CRB}(\hat{\theta}_j;\mathbf{R}_x) \le \rho_j, \ \forall j \in \mathcal{J}
\end{aligned}
$$

>power budget for artificial noise